

## Research Interest

I am interested in exploring Wearable and On-device Agents. I believe that for AI to truly mirror human perception, it must move toward multisensory intelligence—leveraging sensors and the "five senses" captured directly from the physical world. Furthermore, I am convinced that agents capable of integrating our digital and physical worlds will become increasingly vital, with embedded agents playing a central role in this evolution.

## Education

**New York University** Jul 2026 – present

PhD in Computer Science (*Advisor*: Muhammad Shafique)

*Area*: Embedded AI, Wearable & On-device Agent, Multisensory intelligence

**Korea Advanced Institute of Science and Technology** Sep 2021 – Feb 2024

MS in Artificial Intelligence (*Advisor*: Minjoon Seo)

*Area*: Natural Language Processing, Machine Learning, Information retrieval

**SungKyunKwan University** Mar 2014 - Aug 2020

BS in Data Science & BA in Child and Adolescent Education

*Cumulative GPA*: 4.17/4.5

2 years leave for Military service (Sep.2015-Sep.2017)

## Experience

**Mila @ Montreal, Canada** June 2025 - Mar 2026

*Visiting Researcher (Advisor: Siva Reddy)*

I worked on understanding how current Vision Language Models (VLM) are good at tackling knowledge-intensive question in the wild setting [10].

**SoftlyAI @ Seoul, Korea** May 2024 - May 2025

*AI research Engineer*

SoftlyAI is a startup working on the AI Agent for the professionals. Our team worked on building conversational agentic RAG engines that powers the company's core products (clinic coordination, finance assistant).

**LG AI @ Seoul, Korea** Sep 2023 - Feb 2024

*Research Intern*

I worked on an evaluation benchmark, InstructIR, for measuring user-aligned instance-wise instruction following ability of information retrieval at Language lab [7].

**Kakao Enterprise @ Seongnam, Korea** Jul 2020 - Oct 2020

*Machine Learning Engineer Intern*

I worked on knowledge graph embedding (KGE) to better represent complex relationship of large-scale in-house graph data. I also contributed to analyze large scale graph data with distributed system with Hadoop and Spark.

**NAVER @ Seongnam, Korea** Mar 2020 - Jun 2020

*Software Engineer Intern*

I worked on tools for entity linking engine and neural entity linking model for the search system.

**Undergraduate Research Program, KOFAC @ Seoul, Korea** Jan 2019 - Dec 2019

*Undergraduate Researcher*

I worked on a project for building a fake news detection system for news. Our team is grouped in multidisciplinary

with social science, media, and industrial engineering major.

**Republic of Korea Air Force @ Jinju, Korea**

Sep 2015 - Sep 2017

*Sergeant*

I served as a training assistant for the Republic of Korea Air Force Training Wing.

## Honors and Awards

- Best paper, NAACL 2025 [8]
- 1st Place, VALUE Challenge Retrieval Track, ICCV 2021 [1]
- Academic Excellence Scholarship, Sungkyunkwan University, Korea (2019)

## Selected Papers

- [11] Myeong Seok Oh, Dong-Yun Kim, **Hanseok Oh**, Chaeon Kang, Joeun Kang, Xiaonan Wang, Hyunjung Park, Young Cheol Jung, and Hansaem Kim. “Subject-level Inference for Realistic Text Anonymization Evaluation”. In: *ACL*. 2026.
- [10] **Hanseok Oh**, Parishad BehnamGhader, Benno Krojer, Hyunji Lee, Paul Liang, Siva Reddy, and Verna Dankers. “CRAG-MM-Diagnostics: Enabling Stage-Wise Analysis of Knowledge-Intensive VQA”. In: *ECCV*. 2026.
- [9] Hyunji Lee, Seunghyun Yoon, Yunjae Won, **Hanseok Oh**, Geewook Kim, Trung Bui, Franck Dernoncourt, Elias Stengel-Eskin, Mohit Bansal, and Minjoon Seo. “Instruction Tuning with and without Context: Behavioral Shifts and Downstream Impact”. In: *EACL*. 2026.
- [8] Seungone Kim, Juyoung Suk, Ji Yong Cho, Shayne Longpre, Chaeun Kim, Dongkeun Yoon, Guijin Son, Yejin Cho, Sheikh Shafayat, Jinheon Baek, Suehyun Park, Hyeonbin Hwang, Jinkyung Jo, Hyowon Cho, Haebin Shin, Seongyun Lee, **Hanseok Oh**, Noah Lee, Namgyu Ho, SeJune Joo, Miyoung Ko, Yoonjoo Lee, Hyungjoo Chae, Jamin Shin, Joel Jang, Seonghyeon Ye, Bill Yuchen Lin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. “The BiGGen Bench: A Principled Benchmark for Fine-grained Evaluation of Language Models with Language Models”. In: *NAACL*. 2025. **Best Paper Award**.
- [7] **Hanseok Oh**, Hyunji Lee, Seonghyeon Ye, Haebin Shin, Hansol Jang, Changwook Jun, and Minjoon Seo. “INSTRUCTIR: A Benchmark for Instruction Following of Information Retrieval Models”. In: *ACL workshop on Knowledge-Augmented NLP*. 2024. **Oral**.
- [6] **Hanseok Oh**, Haebin Shin, Miyoung Ko, Hyunji Lee, and Minjoon Seo. “KTRL+ F: Knowledge-Augmented In-Document Search”. In: *NAACL*. 2024.
- [5] Yongrae Jo, Seongyun Lee, Aiden SJ Lee, Hyunji Lee, **Hanseok Oh**, and Minjoon Seo. “Zero-shot dense video captioning by jointly optimizing text and moment”. In: *arXiv preprint*. 2023.
- [4] Hyunji Lee, Jaeyoung Kim, Hyeon Chang, **Hanseok Oh**, Sohee Yang, Vlad Karpukhin, Yi Lu, and Minjoon Seo. “Nonparametric Decoding for Generative Retrieval”. In: *ACL Findings*. 2023.
- [3] **Hanseok Oh**, Seojin Nam, and Yongjun Zhu. “Structured abstract summarization of scientific articles: Summarization using full-text section information”. In: *JASIST* (2023).
- [2] Hyunji Lee, Sohee Yang, **Hanseok Oh**, and Minjoon Seo. “Generative multi-hop retrieval”. In: *EMNLP*. 2022. **Oral**.
- [1] Aiden Seungjoon Lee, **Hanseok Oh**, and Minjoon Seo. “ViSeRet: A simple yet effective approach to moment retrieval via fine-grained video segmentation”. In: *ICCV Workshop on Closing the Loop between Vision and Language (CLVL)*. 2021.

## Services

- Reviewer: ARR, ACL, NAACL, EMNLP
- Sub-reviewer: NeurIPS, ECCV